

TOPIC - AI-Based **Posture Correction** **System**

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Empathy

In today's digital environment, students, programmers, office workers, and gamers spend long hours sitting in front of computers and laptops. During observation and informal interaction with users, it was found that many people unknowingly maintain poor sitting posture while working. Common issues observed included leaning too close to the screen, bending the neck downward, slouching, and maintaining incorrect posture for long periods.

Most users are aware that poor posture affects their health, but they usually become too focused on their work and fail to correct themselves regularly. Over time, this can lead to neck pain, back pain, eye strain, fatigue, and reduced productivity.

Existing posture correction solutions often require wearable devices, expensive hardware, or complex setup procedures. Many users prefer a lightweight and affordable solution that works directly through a laptop webcam without additional equipment.

Based on these observations, the need for a simple real-time posture monitoring system was identified.

Define

Problem Statement

"How might we create a simple and accessible system that can monitor a user's posture in real time using only a webcam and provide alerts whenever poor posture is detected?"

Key Problems Identified-

- Users sit too close to screens.
- Users bend their neck downward while working.
- People are unaware of bad posture habits.
- Existing posture correction systems are expensive or inconvenient.
- Long durations of poor posture can negatively affect health and productivity.

Objectives

- Detect face position using computer vision.

- Monitor posture continuously through a webcam.
- Identify bad posture conditions in real time.
- Display warning alerts for poor posture.
- Create a lightweight and user-friendly application.

Ideate

During the ideation phase, multiple approaches for solving the posture monitoring problem were considered.

Ideas Generated

1. Wearable posture correction device
2. Mobile application with posture tracking
3. AI-based body pose estimation system
4. Webcam-based posture correction system using computer vision

After comparing different ideas, the webcam-based computer vision solution was selected because:

- it does not require extra hardware,
- it is cost-effective,
- it is easier to use,
- and most laptops already contain webcams.

Initially, advanced machine learning and pose estimation techniques were considered. However, for simplicity and efficiency, a lightweight OpenCV-based face detection system was chosen for the prototype.

Final Idea

The final system:

- captures webcam input,
- detects the user's face,

- calibrates a normal sitting posture,
- monitors face movement and position,
- identifies bad posture,
- and displays real-time alerts and posture score updates.

Prototyping

A working prototype of the AI-Based Posture Correction System was developed using Python and OpenCV.

Tools and Technologies Used

- Python
- OpenCV
- Streamlit
- Streamlit WebRTC
- NumPy

Prototype Features

- Real-time webcam access
- Browser-based interface
- Face detection using OpenCV Haar Cascade
- Automatic posture calibration
- Detection of poor posture
- Warning messages
- Posture score display

Working of the Prototype

1. The webcam captures live video frames.
2. OpenCV detects the user's face in each frame.

3. The system stores the initial face position as the baseline posture.
4. If the face moves significantly downward or closer to the camera:
 - the system detects bad posture.
5. Warning messages are displayed after continuous bad posture.
6. The posture score decreases gradually when poor posture continues.

The prototype was tested locally and later adapted for browser-based deployment using Streamlit WebRTC.

Testing

The prototype was tested under different sitting conditions to evaluate its performance.

Test Cases

- Proper sitting posture
- Leaning too close to the screen
- Looking downward continuously
- Moving away from the camera
- Different lighting conditions

Observations

- The system successfully detected face movement and posture changes.
- Real-time feedback helped users become aware of incorrect posture.
- Warning messages improved user interaction.
- Browser webcam support improved accessibility.

Limitations Identified

- The system currently uses only face detection instead of full-body tracking.
- Detection accuracy depends on lighting conditions.
- Extreme face angles may reduce detection performance.

Future Improvements

- Full body posture estimation
- Audio warning alerts
- Mobile support
- Posture analytics dashboard
- AI-based posture classification

Conclusion-

The AI-Based Posture Correction System successfully demonstrates how computer vision can be used to solve real-world health-related problems in a simple and practical manner. Through the EDIPT Design Thinking process, a lightweight and user-friendly posture monitoring solution was designed, prototyped, and tested successfully